

10-Mark Questions & Answers — MPCA Module 4: Memory & Peripherals Interfacing

Q1. Explain Memory Interfacing with 8086 (RAM/ROM decoding), 8255 PPI, 8257 DMAC, and 8259 PIC in detail with block diagrams and control words.

Comprehensive 10-mark master answer covering:

1. 8086 Memory Banking (Even/Odd banks, A0, /BHE) and 74LS138 address decoding.
2. 8255 PPI Architecture, Control Word format, and Mode 0/1/2 operations.
3. 8257 DMA Controller operation, HOLD/HLDA timing, and Burst/Single modes.
4. 8259 PIC Block Diagram, IRR/ISR/IMR registers, vector transfer, and cascading.